EXPERIMENT - 8

User Management and Role-Based Access Control in Zoho Creator

Library Book Issue Management

Application Name	Library Book Issue Management
Form Name	Book_Issue_Request_Form
Report Name	Book_Issue_Request_Report
Student Name	CALEB D
Student Roll No.	241401016
Experiment	8
Platform	Zoho Creator
Record Date	09 October 2026

Lab Record	Configuration of Library Admin, Library Staff and Library Member roles and permissions, with Development demo-user tests and the accepted real Staff assignment clearly documented.

Department of Computer Science and Business Systems

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1. Objective

To configure role-based access control so administrators manage library data, staff process requests, and members submit and view only permitted records.

2. User Groups and Responsibilities

Role	Responsibility
Library Admin	Manage library records, reports and dashboard; application owner retains design and account administration.
Library Staff	Review requests, update processing status and notes, and use reports and dashboard.
Library Member	Submit a book issue request and view permitted own records without internal staff data.

3. Roles and Permission Profiles

Created roles named **Library Admin**, **Library Staff** and **Library Member**, with corresponding same-named permission profiles. The role hierarchy is the existing CEO role followed by Library Admin, Library Staff and Library Member. Peer sharing is off.

A role establishes the position in the hierarchy; a permission profile controls the permitted application components and actions. The owner remains Super Admin. A business Member_Type value such as Student or Faculty does not itself grant an access-control role.

4. Record and Field Protection

Members do not have View All or Modify All access. Staff_Notes and Confirmation_Sent are hidden for Members, and Request_Status is read-only. Staff can process records but cannot directly edit Confirmation_Sent. Dashboard access is limited to Admin and Staff profiles.

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5. Permission Matrix

Capability	Library Admin	Library Staff	Library Member
Submit request	Allowed	Allowed	Allowed
Main report	All library records	All library records	Permitted own records
Pending / Available reports	Allowed	Allowed	Not granted
Edit requests	Allowed	Allowed	Not granted
Delete requests	Allowed	Not granted	Not granted
Dashboard	Allowed	Allowed	Not granted
Request_Status	Manage	Manage	Read-only
Staff_Notes	Visible / editable	Visible / editable	Hidden
Confirmation_Sent	Administrative visibility	Read-only	Hidden
Import / export / API	Administrative profile	Not granted	Not granted
Design / account admin	Owner-managed separately	Not granted	Not granted

6. Configuration Procedure

1. Created the three library roles and set the hierarchy with peer sharing disabled.
2. Created matching permission profiles and assigned form, report and page permissions.
3. Restricted Members to own records and removed edit, delete and dashboard access.
4. Applied field-level restrictions to staff notes and confirmation tracking.
5. Configured four Development demo users for Admin, Staff and two separate Member simulations.
6. Tested record visibility, allowed controls, restricted fields and status initialization.
7. Added the real Hariharan account with Library Staff role/profile; the invitation was accepted.

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7. Development Role-Test Results

These tests used Zoho Creator Development **View as** demo users. They validate simulated role behavior; they are not separate-account credential login tests.

Identity / test	Observed result	Status
demouser_1 - Member	Submitted Ananya; saw only own request, not owner or second-member records.	Pass
demouser_4 - Member	Initially saw no data; submitted Vikram and saw only that Pending request.	Pass
Member restrictions	No Edit/Delete controls; internal notes and confirmation flag absent.	Pass
Member dashboard	Navigation excluded dashboard; direct attempt did not expose dashboard content.	Pass
demouser_2 - Staff	Viewed all records; saved Ananya as Under Review with internal notes.	Pass
Staff restrictions	Confirmation_Sent disabled; no Delete control.	Pass
Staff dashboard / queue	Five total records; two Pending; queue showed Aarav and Vikram.	Pass
demouser_3 - Admin	All five records, dashboard and Edit/Delete controls visible. Delete was not executed.	Pass
Member initialization fix	New saved record initialized to Pending by Set_New_Request_Status.	Pass

8. Real-Account Assignment Status

Hariharan was assigned **Library Staff** role and **Library Staff** permission for this application. His invitation acceptance was confirmed in the Users screen. No developer or administrator access was assigned.

Remaining: real separate-account Staff and Member login tests have not yet been executed. The browser remains signed in as Caleb; invitation acceptance alone does not prove effective access under Hariharan's login.

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9 - Implementation Evidence

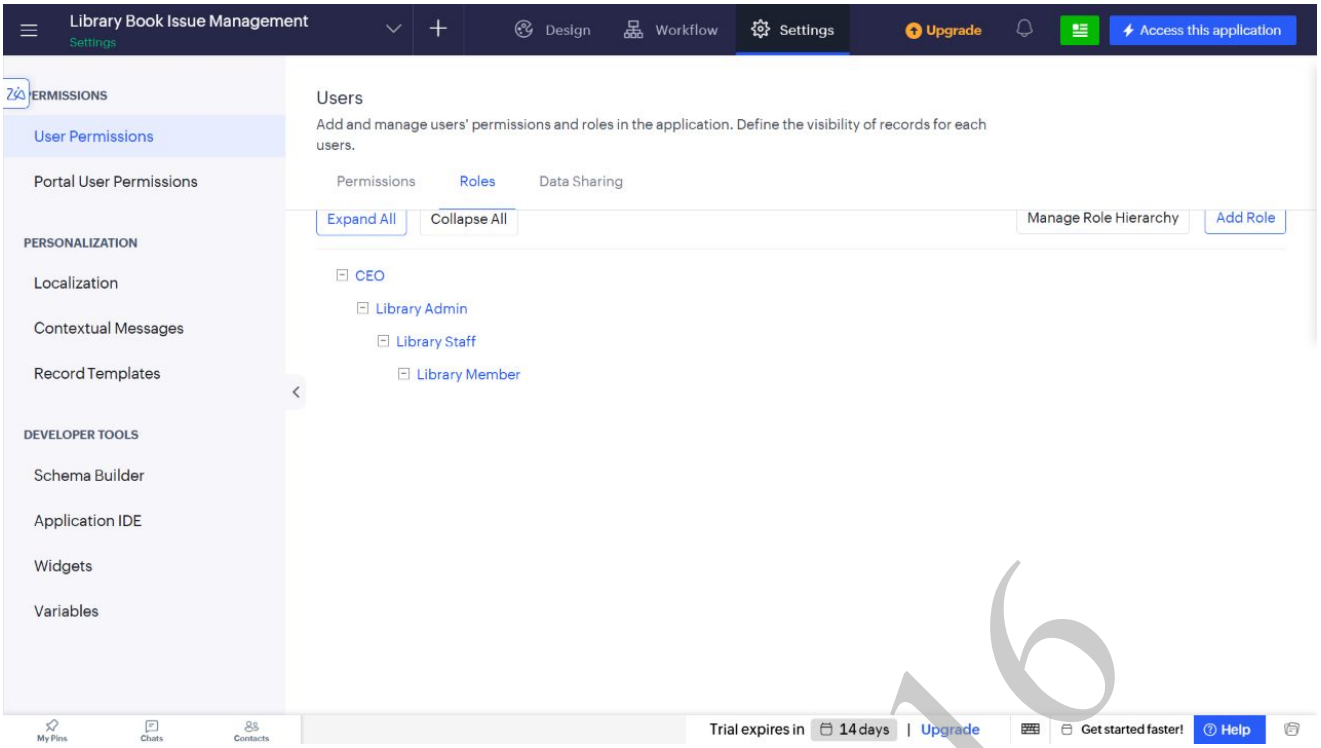


Figure 8.1. Configured library role hierarchy in the application.

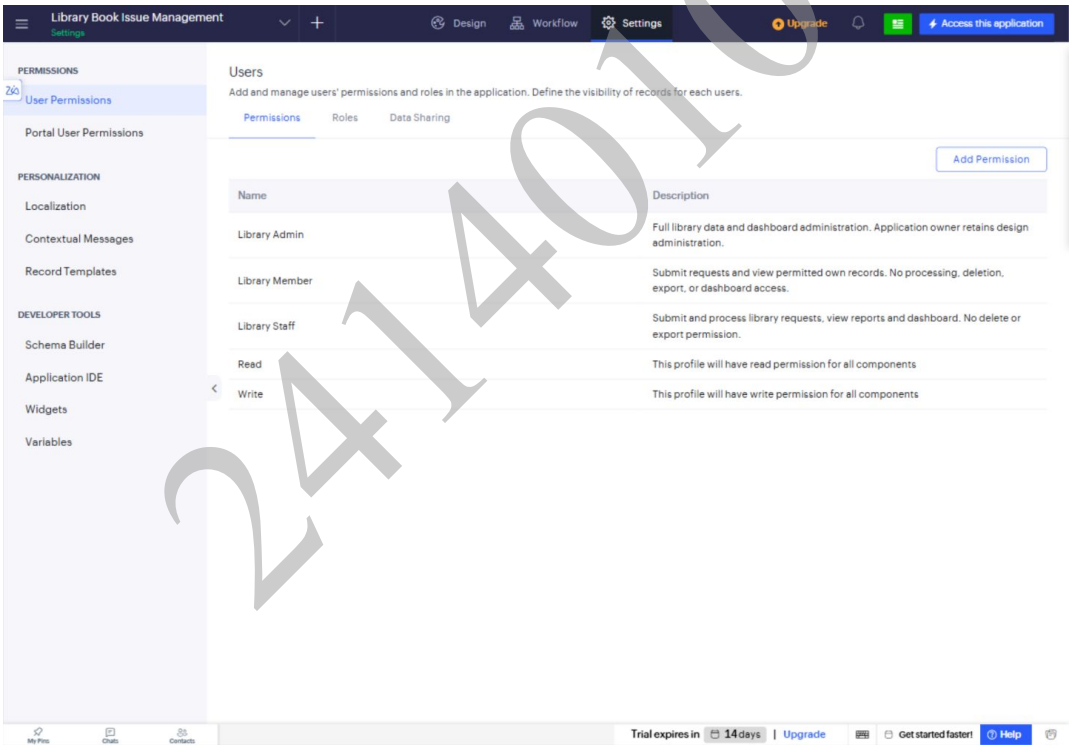


Figure 8.2. Permission profiles for Library Admin, Library Staff and Library Member.

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9 - Implementation Evidence

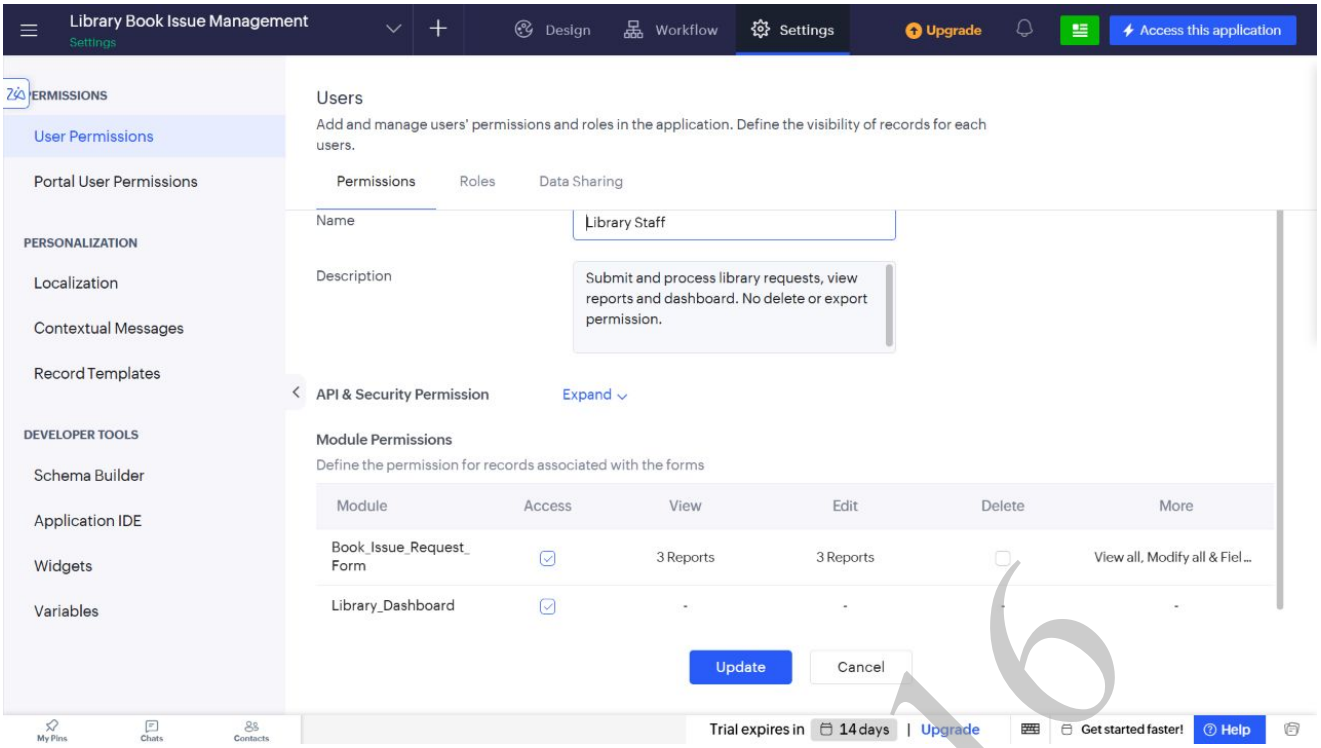


Figure 8.3. Staff report/action permissions used for request processing.

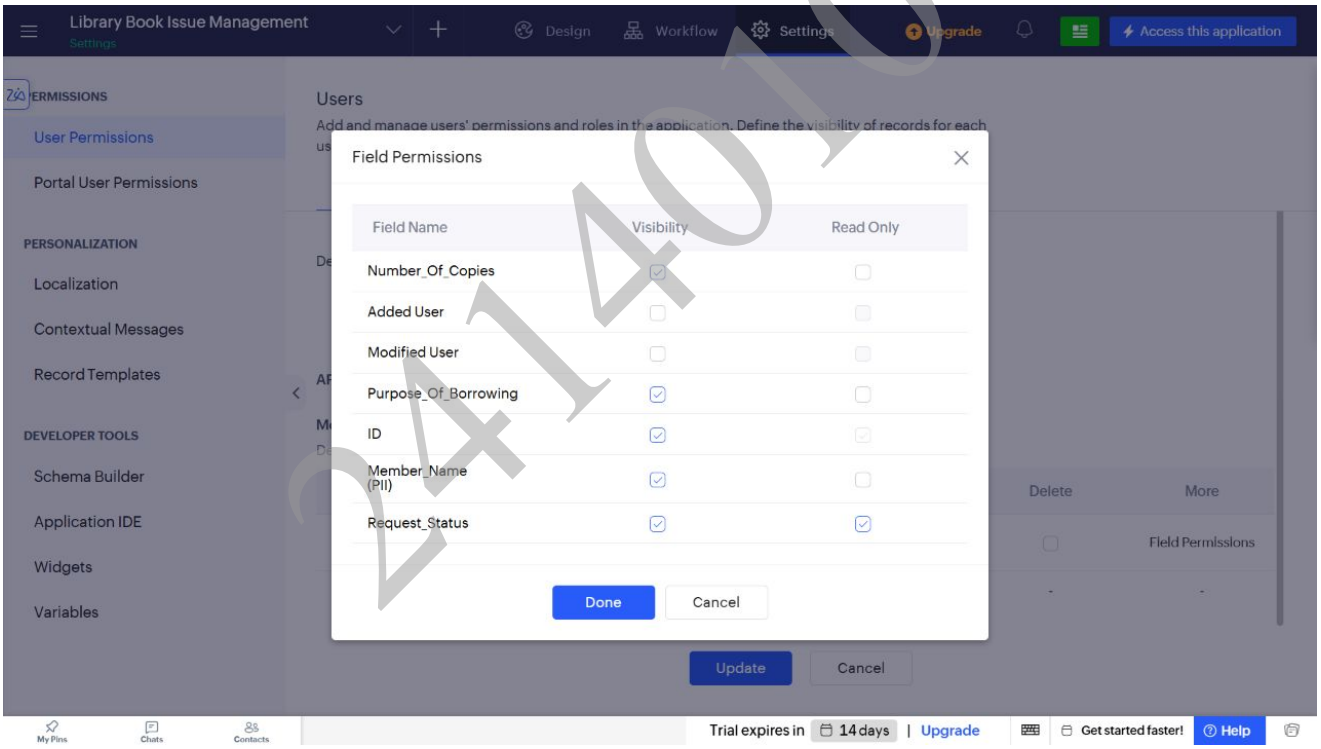


Figure 8.4. Member field permissions protecting staff notes and confirmation tracking.

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9 - Implementation Evidence

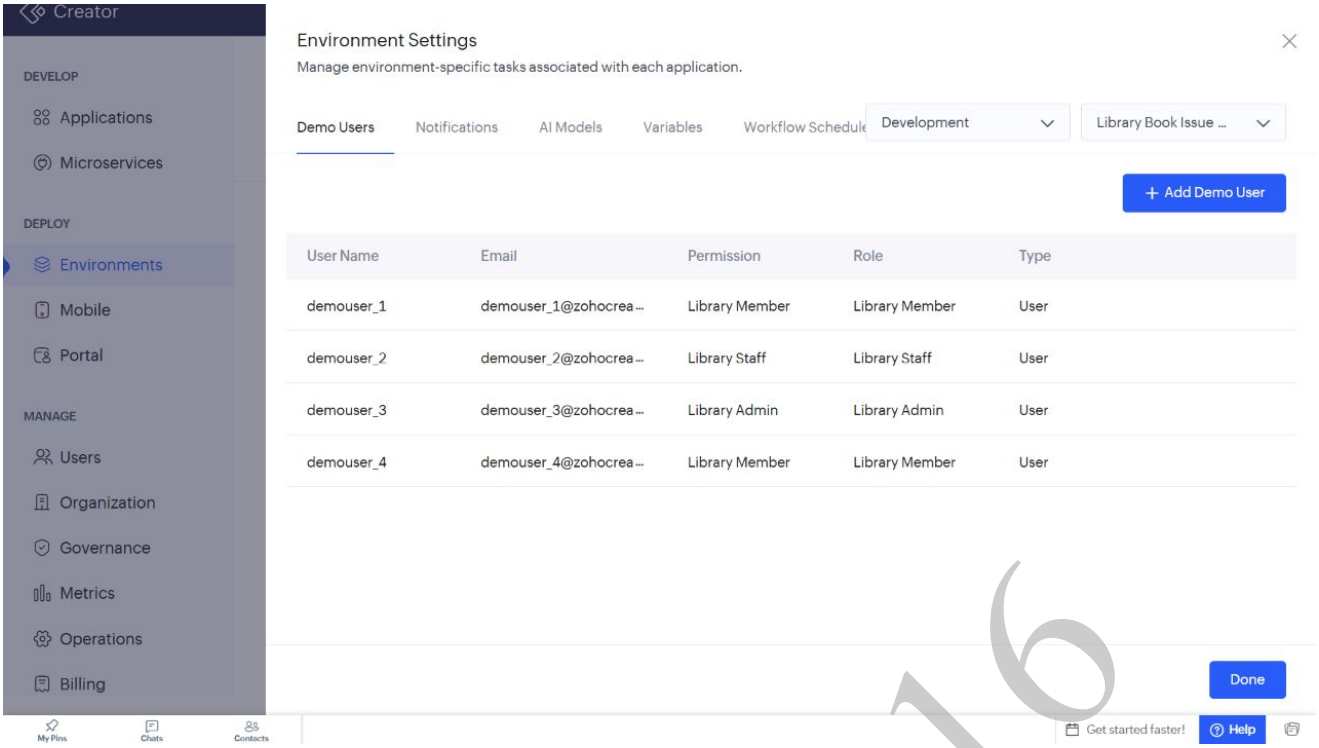


Figure 8.5. Development demo-user assignments used for simulated Admin, Staff and two Member tests.

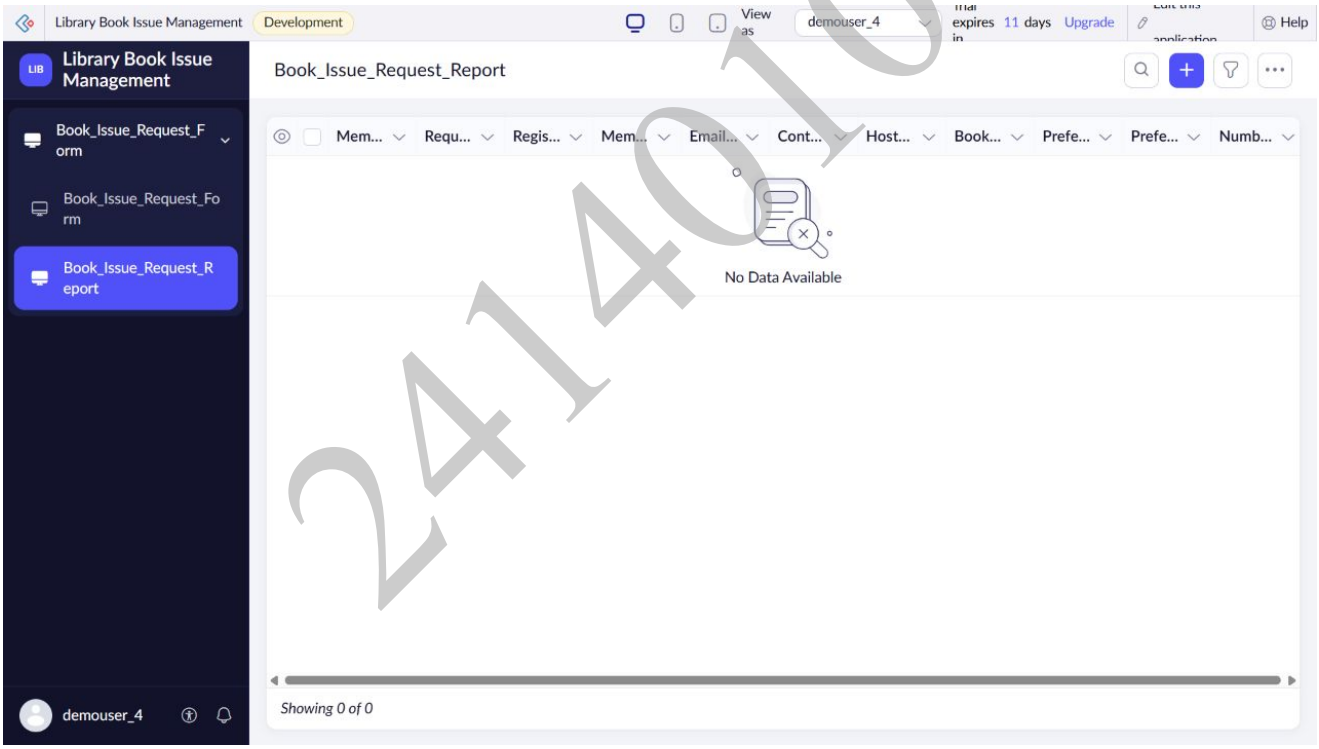


Figure 8.6. Second Member simulation sees only its permitted record; this is View as testing.

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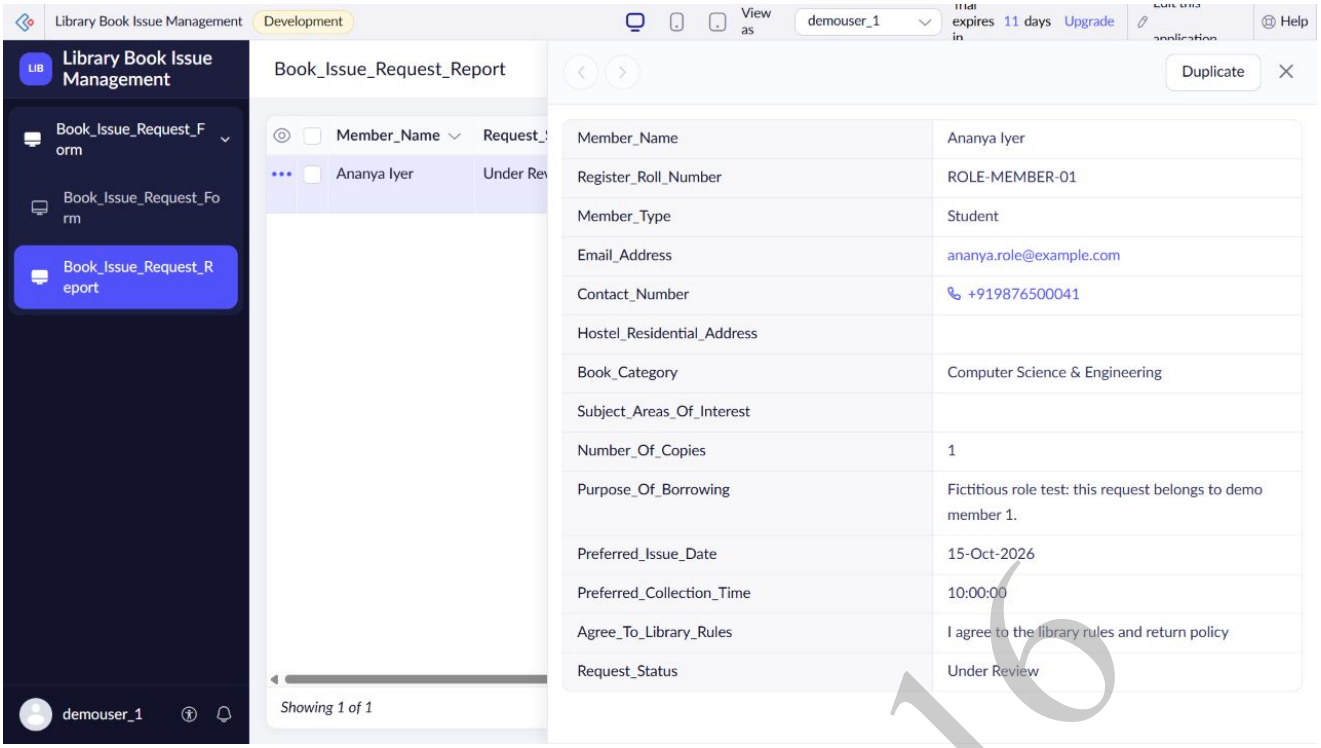


Figure 8.7. Member simulation sees reviewed status while internal processing information remains hidden.

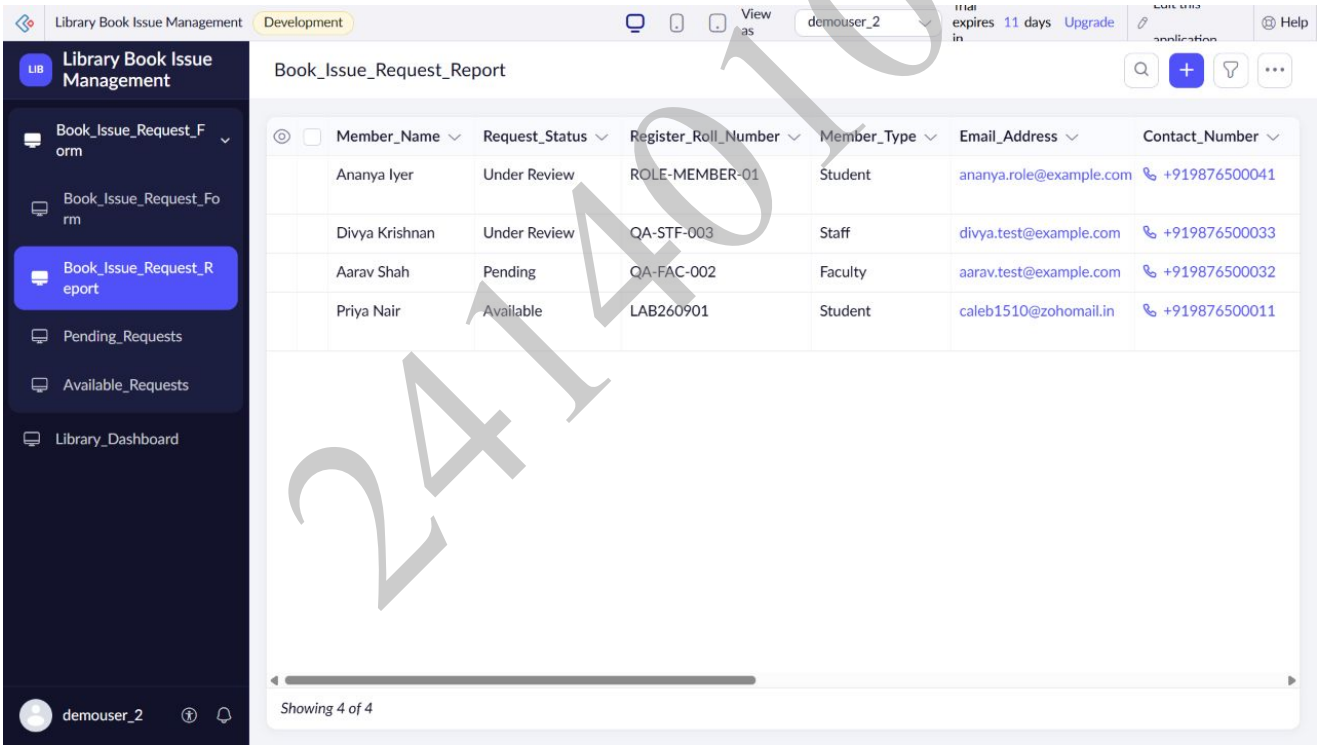


Figure 8.8. Staff simulation sees all requests and performs request processing.

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9 - Implementation Evidence

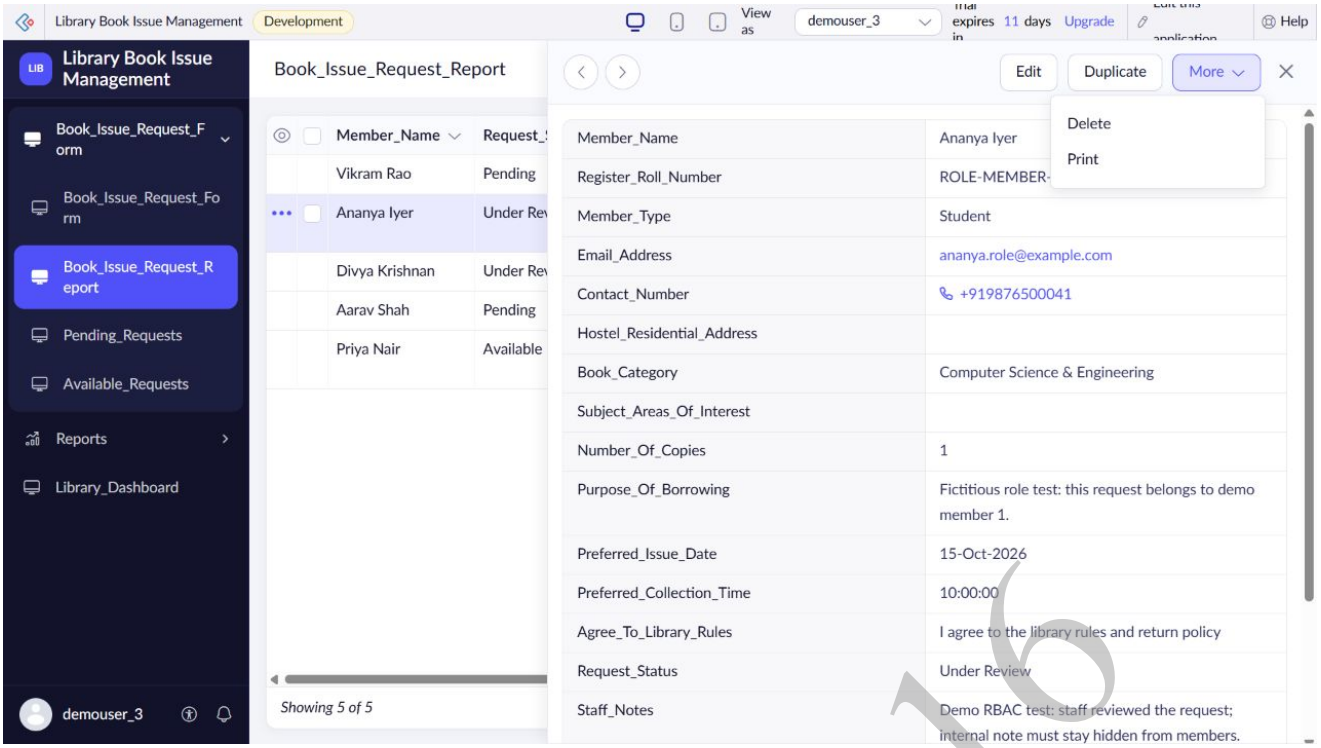


Figure 8.9. Admin simulation exposes administrative record controls; deletion was inspected, not executed.

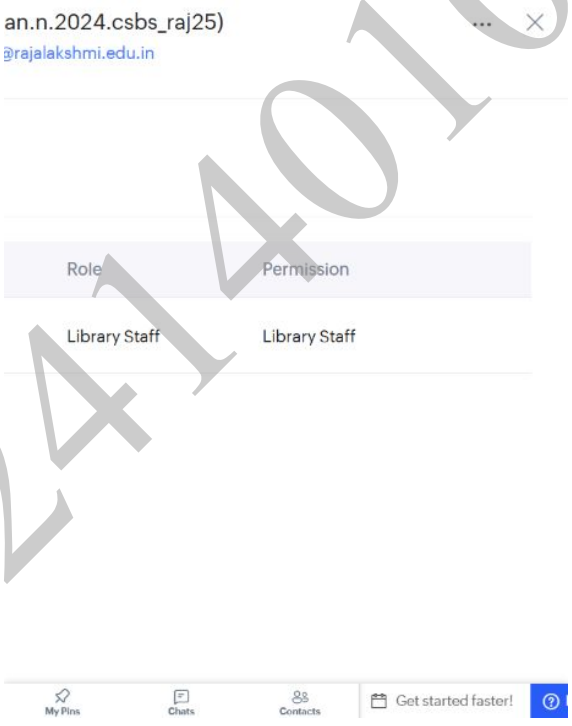


Figure 8.10. Accepted real Hariharan account assigned Library Staff role and permission. This proves assignment, not a login test.

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10. Procedure for Remaining Real-Login Test

1. Have Hariharan sign in privately with his own Zoho credentials and open the Production application. Passwords and MFA must be entered by the account holder.

2. Confirm the visible signed-in identity, then check Staff reports, dashboard, allowed edits, disabled confirmation field and absence of Delete.

3. Create a clearly fictitious request using an example.com address; avoid sending an external email.

4. Return to the owner account and assign the tester the Library Member role and permission profile for the Member test.

5. Sign in again as the tester and confirm own-record visibility, no access to owner records, no dashboard or editing, and hidden internal notes.

6. Submit a Member request, verify Pending status, capture actual-account evidence and record the final intended assignment.

11. Expected Learning Outcomes

- Distinguish roles, permission profiles and business member categories.

- Apply least-privilege access to forms, reports and internal fields.

- Verify record isolation using two Member simulations.

- Distinguish demo-user simulation, invitation acceptance and actual separate-account login testing.

12. Result

Three library roles and three matching permission profiles were configured. Development demo-user tests passed for the documented access checks. The real Staff invitation was accepted; actual separate-account Staff and Member login verification remains pending.

Conclusion

The configured access model separates administration, request processing and member submission. The evidence demonstrates configuration and simulated enforcement while explicitly preserving the remaining real-login verification step.

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